

Project plan

Big Chemistry

Big Chemistry *Eindhoven*

<< You write the project document as an agreement between you and your client. In this you record agreements regarding the assignment and you show that you understand what the request is from your client. You also indicate how you are going to approach the assignment, why you choose it and what the end result is.

You can use this template for all projects – chapters or parts that are not applicable to the specific project (eg the section Testing and Configuration Management for non-software assignments) can be omitted.

Texts in <<italic>> are for informational purposes what should be in each section and should be removed from the final document

It is important that this document looks neat and tidy. This means at least that it is error-free and that the layout looks as usual within your IT direction (eg media design). The project document partly determines the first impression, so pay attention to that. Also, do not write unnecessary details in the document.

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Communication

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1. Project Assignment

1.1 Context

<< Briefly describe the company and the context of the assignment. Provide information about the products and services of the company that your assignment focuses on. If you work for an external client of your client – think of a client of an internship company – describe them in the same way. In addition, indicate the concrete reason for the assignment and what developments are taking place in the company or the market that lead to the assignment.>>

Fontys, along with many other universities all across the Netherlands, are working on “Big Chemistry” project. That entails the manufacturing of a fully self driving laboratory (SDL).

1.2 Goal of the project

<<Describe the goal of the project here. Think of:

What is the problem that needs to be solved or what is the opportunity that needs to be used?

What does the desired situation look like?

What benefits does the project offer?

What possibilities (capabilities, facilities) does the product or project result offer?

>>

<<NB To make your goals as concrete as possible, you should already have a good idea of the problem. What exactly is the issue? What is the problem to be solved or what is the challenge? Why is this question there? What is the urgency? What caused it? What are the consequences if nothing is done? And what has already been done to arrive at an answer? It is essential that you look critically at the client's needs. Is the problem outlined actually the problem? And is your client's question actually the right solution? Ask critical questions and try to arrive at the correct problem statement together with the client. If more research is needed to determine this, include this in your approach.>>

Fontys, along with TU/e, are working on building a demonstrator, which involves sample preparation with a pipetting robot, sample transport with a robot arm and absorption measurements with a plate reader, all working under SILA2 standard. The final goal of the whole project is to create a fully autonomous chemistry laboratory environment to aid scientist in their research.

1.3 The assignment

<<Formulate the assignment. The assignment definition itself should consist of a text that is as short and concise as possible in which the assignment is clearly formulated. What are the specific requirements/wishes of the client at the start of the project? What are the minimum (quality) requirements that the end result must meet? You may already provide a list of functional and non-functional requirements for the end product as an attachment>>

The assignment consists in the continuation of a self-developed API protocol for controlling the Ufactory Lite 6 robot arm. The arm should be able to execute its tasks without making collisions with the other equipment in the lab. The arm must be able to walk through its trajectory unimpeded. The demonstration includes the arm calibrating itself and pick up the small plastic boxes, where the solutions would be placed. The Technologies involved are *trajectory planning, inverse kinematics, Inter-machine Communication, ROS2 framework, Python, Digital Twin*

Functional Requirements:

- *The arm must be able to position itself(calibration).*
- *The arm must NOT collide with surrounding equipment or itself(over-reaching).*
- *The arm must be able to find its trajectory.*
- *The arm must pick the small container up and put it on its destination.*

Non-functional Requirements

- *A conclusion towards tooling. Should the default tooling from Ufactory or ROS framework be used?*
- *Conclude if it is beneficial to use ROS2's MoveIT library for trajectory planning(Part of ROS framework implementation experiments).*
- *Conclude what calibration implementation would be the most beneficial. QR-target-based or something else.*
- *Version control with Git.*
- *Research topics with DOT.*

1.4 Scope

<< Indicate the scope. If necessary, make a context diagram for clarification that shows the relationships with other systems and the environment. This section should also describe what will not be delivered. For example, if you agree to deliver a high fidelity prototype, then (part of) the implementation and management falls outside your scope. Make this as concrete as possible so that there are no misunderstandings between you and your client. >>.

The project includes:	The project does not include:
1 Research on ROS2 and similar technology	1 Explicit development on SILA2
2 Decision and consultation with backed-up evicence on which techonolgy is better-suited for the task at hand.	2 Works on connecting the other equipment
3 Implementation of path trajectory algorithm using either default tooling or ROS2	
4 Demonstration of the robotarm's capabilities to safely deliver payload from point A to B.	

1.5 Conditions

<< Indicate, where necessary, what the preconditions are. For example, consider technology set by the company. Note that a critical attitude remains important here!>>

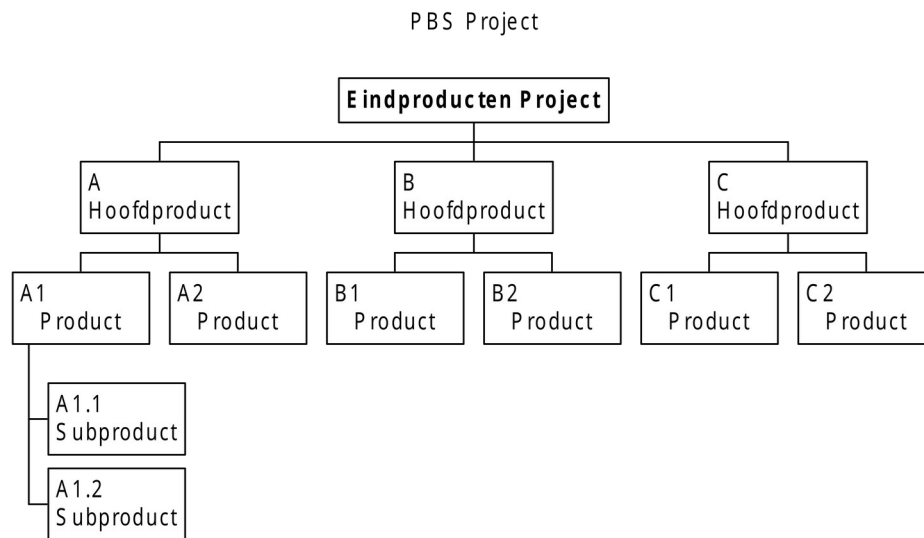
The company is under consideration on which technology to apply for the remainder of the project. The diversion in opinions is between the default tooling provided by Ufactory and it is up to the intern (me.) to experiment and reach a conclusion.

1.6 Finished products

<<A Product Breakdown Structure of the end and intermediate products that the project will deliver with a short description in text of each product. The end products are more than the project plan and the product itself. Also, for example, requirements and architecture documents and research and test reports are typical parts of a PBS. These documents are important for the relevant stakeholders during development as well as during the transfer and during the management phase. During the project you can change the PBS and you can add or remove products in consultation. >>

Additionally the products to be presented are as follows:

- Research paper on the technologies mentioned earlier with a final conclusion drawn at the end.
- Demonstation on the capabilites of the Ufactory Lite 6 robot arm using the selected technology(video and live presentation format).
- Project plan.
- Portfolio.
- Git repository with source code and relevant documentation.



1.7 Research questions

<<Describe the most important research questions you want to answer during your internship. Define a main question with sub-questions derived from it. Keep in mind that you will be doing investigative work during your entire internship, and that your questions will therefore concern your entire trajectory. During your internship/graduation, more research questions may be of interest and others may turn out to be less relevant. Describe only the key research questions that will have the greatest impact on your project. Other research aspects can be elaborated in more detail during your internship and can then be explained with a short substantiation (in your portfolio or orally). >>

Could it be beneficial to use ROS2 instead of the current built-in tooling provided by Ufactory?

- *What are the physical limits of the hardware?*
- *What and how the default tooling from Ufactory functions?*
- *What is the ROS2 framework?*
- *How do the two technologies compare?*
- *How is software developed using the default tool-set?*
- *How is software developed using the ROS2 framework?*
- *What are the differences in embedded development between ROS2 and default tools?*

How would a path trajectory algorithm be implemented to the Ufactory Lite 6 model?

- *What is trajectory planning?*
- *How would such algorithm be implemented using default tools?*
- *How would such algorithm be implemented with ROS2 framework?*
- *What would be considered a good demonstration of the algorithm? What would be an acceptable criteria?*

2. Approach and Planning

2.1 Approach

<< Indicate here which method you follow in your project plan, for example whether you use a waterfall or scrum method. Also indicate how you will approach the problem definition phase and completion phase. With a scrum approach you can think of length of sprints, set-up of your sprints, stand-up, set-up of demos, retrospective, etc.)>>.

It is concluded that for this project the scrum method will be followed. On a personal and non-personal level. It is logical, since the company already implements it in their business with daily stand-ups. Below I have indicated approach elements and how I will be dealing with them.

Organization and planning

- Planning is done following the Canvas course and how the weeks are separated. For example:
Week 1-2: Project plan, meaning that the Project Plan must be done by the end of week 2.

Tracking

- Tracking tasks is done via a calendar agenda

Sprint set-up

- Each sprint will be around 2 weeks. By the end of each sprint the next one will be planned based on the context of the previous.

Stand-ups

- There are daily stand ups for the whole project arranged by the company. Additional meetings with individuals will be arranged based on "need" basis. For example: Mentor meetings are arranged every Monday and Wednesday.

Retrospective and reflections

- Sprint retrospectives are done at the end of each sprint, concurrent with the planning of the next one.
- Reflections and feedback will be sought from peers, assessor and company mentor either in written or spoken form(emails, teams messages or face-to-face meetings).

2.1.1 Test approach

Testing will be done using the following methods:

- **Unit testing**
- **Simulation**

During the course of the sprints, testing and experiments will be conducted when new features are done, to better determine compatibility with previous code and/or remove bugs.

2.2 Research methods

<< Describe (per research question and for the entire project) which methods (see [ictresearchmethods.nl](#) or [cmdmethods.nl](#)) you will use to answer the most important questions within your project (= how you will substantiate the most important choices). Do not only mention the method, but also briefly explain how you will use it (e.g. who will you interview and for what purpose?). Of course, your approach can still be adjusted during your internship.>>

The research method used for the project will be the DOT framework. Each main and subquestion will be researched based on that method. There are no specific restrictions when it comes to gathering necessary information.

I shall not outline each specific method since each question requires unique attention on what the best approach is. A common ground for each question would be the **What and Why** approach with the rest (**Lab, Library**) catering to specific situations. The agile methods demands flexibility and to keep that I shall not limit myself to specific methods will be unproductive. Thus if I determine that a method would not fit the specific case it will be removed. In my approach I want to keep flexibility to maximum as to be able to easily adapt and adjust to the current situation.

2.3 Learning outcomes

<<Discuss how you are going to demonstrate each learning outcome in the project. The easiest way is to think about which of the professional products you are going to use as evidence for each of the learning outcomes.>>.

The learning outcomes are as follows:

Professional Duties – I will fulfill this outcome by showing my work to my superiors during the internship and get feedback from them. In the end I would like to present my research, code and demo the functionality I've built.

Situation-orientation – I will apply all I have learned from my previous semester, such as: research with DOT framework, agile scrum management, SOLID coding standards and proactive attitude with active self reflection.

Professional Standard – I will demonstrate this learning outcome by meticulously and actively participate in day-to-day activities in the company. Diligently and responsibly researching and implementing ideas in parallel with advice from mentor and peers. Completing task in the agenda and keeping myself accountable will be the core of this learning outcome. Supporting documents will be created as the process goes on.

Personal Leadership – I will pay attention to my learning ability and seek improvement. Feedback from peers and superiors will be documented in feedpulses. Additional feedback documents, if needed, will be created to compile all the feedback received.

2.4 Breakdown of the project

<<Show the rough breakdown in phases or sprints of the project here.>>

Phase 1: Familiarization with the default tool-set. Testing the capabilities of the hardware. Running custom scripts for testing.

Phase 2: Research and implementation of ROS2 framework. Testing custom instructions and comparison with the default toolset.

Phase 3: Research on path-finding algorithm on default tool-set and ROS2. Comparison and final choice for technology framework with backed-up research and evidence.

Phase 4: Finalizing a demo and compiling all concurrent research documents, ready to be presented to faculty.

2.5 Time plan

<< Depending on your project method, you will be able to work out the phasing in more or less detail. Below is a possible table that you can use for this.

Note that with an agile approach, most projects still have a problem analysis/orientation phase (or 'sprint 0'), as well as a completion/evaluation phase.

Also make sure that you reserve enough time for your portfolio and start on time.>>

Phasing	Effort	Start	Ready
1 Familiarization phase	minimal-middling	4 th March	25 th -31 st March
2 Research and implementation (ROS and UFactory tool-set)	significant	1 th April	15 th -25 th April
3 Research and implementation on path-finding algorithms and orientation(ROS and Ufactory tool-set).	significant	26 th April	25 th May
4 Finalization and reflection phase	middling	26 th May	16 th June

* Dates are just for orientation and are subject to change

3. Project Organization

3.1 Team members

<< Describe the organization of the project with its immediate environment. An organization chart can be displayed for clarification. Indicate in descriptive form which roles are included in the organization chart with the associated authorities and responsibilities. It must be clear who is authorized to do what and what can be expected of whom. Indicate who is involved in your project and what his/her function is and what the role is within your project. For example, someone with the function 'manager of department X' can have the role of Product Owner in your project. In this project, both the internship/graduation organization and

Name + Phone + e-mail	Abbr.	Role/tasks	Availability
j.geurts@fontys.nl , 06 53183066		Company mentor, Educator	4 days a week

Fontys are stakeholders. So include your internship teachers and yourself in this schedule.>>

3.2 Communication

<< Indicate which communication/attunements there are. Think of coordination with company supervisor, teacher supervisor and other stakeholders. How and how often do these attunements take place?>>

Majority of the communication between supervisors are using the Teams platform. Additionally, regular meetings with mentor and assessor are planned on weekly basis, and are face-to-face at minimum once a week.

3.3 Test environment

<<omit this section if not applicable>>

3.4 Configuration management

The GIT structure will follow the already established. A branch will be created that, which in turn, may branch out to others as features are introduced. This is done to ensure safety and to keep the code-base clean.

projects.fhict.nl/htes-lectoraat/big-chemistry-robotlab/sdl-demonstrator/hardware/robotarm

HTES Lectoraat / Big Chemistry - Robotlab / SDL Demonstrator / hardware / RobotArm

Search or go to...

Project

RobotArm

Pinned

Issues1

Merge requests0

Manage

Plan

Code

Build

Secure

Deploy

Operate

Monitor

Analyze

initial blockly

i888461 authored 3 months ago

3fc6e937

History

Name	Last commit	Last update
feature_definitions	Initial Blockly app sila server and sim...	3 months ago
tests/features	Initial Blockly app sila server and sim...	3 months ago
xarm_sila_server	Initial Blockly app sila server and sim...	3 months ago
.dockerignore	Initial blockly	3 months ago
.gitignore	Initial blockly	3 months ago
CONTRIBUTING.md	Initial Blockly app sila server and sim...	3 months ago
Dockerfile	Initial Blockly app sila server and sim...	3 months ago
README.md	Initial Blockly app sila server and sim...	3 months ago
docker-compose.yml	Initial Blockly app sila server and sim...	3 months ago
interaction_example.ipynb	Initial Blockly app sila server and sim...	3 months ago
pyproject.toml	Initial Blockly app sila server and sim...	3 months ago

README.md

Project information

5 Commits

2 Branches

0 Tags

README

CONTRIBUTING

Auto DevOps enabled

+ Add Wiki

Created on

February 04, 2026

4. Finance and Risks

4.1 Cost budget

<< If specific costs have to be incurred for your project, please indicate these. Think of extra hardware or software investments. Regular matters such as workplace, your internship allowance, etc. do not have to be included.>>
No extra costs are incurred for this project.

4.2 Risks and fall-back activities

<< Define risks. What have you already included in the plan to limit or prevent the risk? What choice is made if the risk does unexpectedly occur? Think of organizational risks (such as the sudden leave of the company supervisor) as well as more substantive risks (for example, what to do if you find out during your internship that it is better for the company to purchase an external application instead of the application to be developed).

Think of real risks that can actually influence your project. For example, there may be a risk that your company supervisor will be absent, for example due to illness or because he is going to do something else. Is there a backup in the company?
>>

Risks are minimal for this project, since by default, a lot of safeguards are implemented, both in hardware and software, any risks listed are unlikely to happen, but are listed just as a precaution

Risk	Prevention activities included in plan	Fall-back Activities
1 Hardware failure	Careful operation of the systems.	Relying on simulations for the time-being.
2 Software brick	Usage of recommended tools by Ufactory.	Relying on simulations
3		

5. Other

<< Describe here everything that is relevant but that you cannot put elsewhere in the document.>>